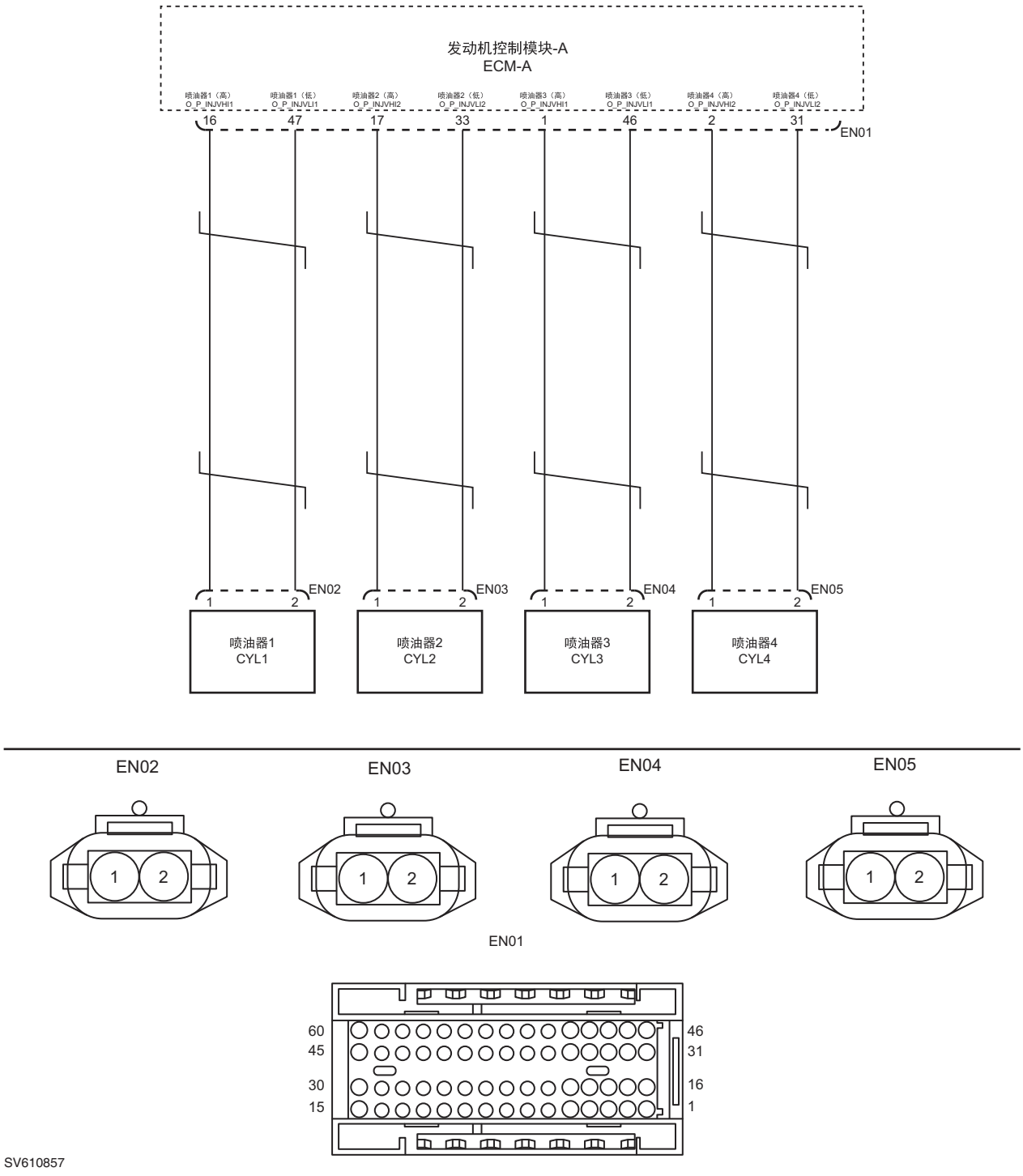
P0201、P0202、P0203、P0204、P0261、P0262、P0264、P0265、P0267、P0268、P0270、P0271、P062D、P062E

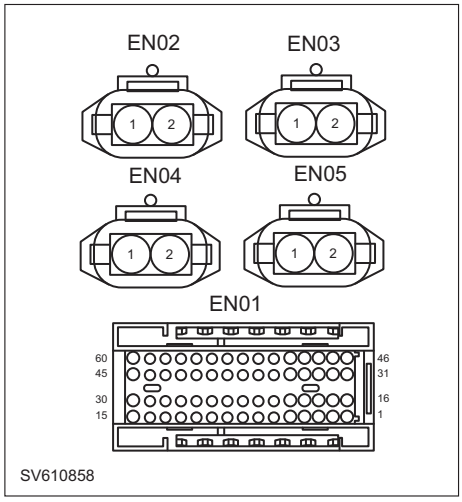
Description of fault diagnosis codes
P0201-13:open load of logical injector 1
P0202-13:open load of logical injector 4
P0203-13:open load of logical injector 2
P0204-13:open load of logical injector 3
P0261-11:short circuit Low Side to High Side logical injector 1
P0262-12:general short circuit logical injector 1
P0264-11:short circuit Low Side to High Side logical injector 4
P0265-12:general short circuit logical injector 4
P0267-11:short circuit Low Side to High Side logical injector 2
P0268-12:general short circuit logical injector 2
P0270-11:short circuit Low Side to High Side logical injector 3
P0271-12:general short circuit logical injector 3
P062D-F0:short circuit of injection bank1
P062E-F0:short circuit of injection bank2
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Line fault of oil atomizer • Oil atomizer fault • Damages of engine control module
Description of circuit diagram
The common-rail oil injector is used to inject very little high-pressure fuel into the combustion chamber exactly. The oil injector is installed on the cylinder cover via taper lock, and copper gasket is used to seal the combustion chamber. The oil injection quantity of common-rail oil injector is controlled by the electronic trigger. The oil injection during start time is controlled by the timing system of electronic diesel engine control system.

Circuit diagram



Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and fuel injector. 4 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON. 3 Diagnose the engine control system with diagnostic apparatus. 4 Are there any other fault codes except for P0201, P0202, P0203, P0204, P0261, P0262, P0264, P0265, P0267, P0268, P0270, P0271, P062D and P062E?
		Yes → Reference: DTC Summary List (EU5).
No ↓		

Step 3	Check the continuity of oil atomizer and ECM harness.



1 Turn the ignition switch to OFF.

2 Disconnect the ECM harness plug.

3 Disconnect the plug of oil atomizer.

4 Measure the resistance between the plug of oil atomizer and ECM plug.

Measuring circuit: the resistance between terminal 1 of EN02 plug and No.16 terminal of EN01 plug.

Measuring circuit: the resistance between terminal 2 of EN02 plug and terminal 47 of EN01 plug.

Measuring circuit: the resistance between terminal 1 of EN03 plug and No.17 terminal of EN01 plug.

Measuring circuit: the resistance between terminal 2 of EN03 plug and terminal 33 of EN01 plug.

Measuring circuit: the resistance between terminal 1 of EN04 plug and terminal 1 of EN01 plug.

Measuring circuit: the resistance between terminal 2 of EN04 plug and terminal 46 of EN01 plug.

Measuring circuit: the resistance between terminal 1 of EN05 plug and terminal 2 of EN01 plug.

Measuring circuit: the resistance between terminal 2 of EN05 plug and terminal 31 of EN01 plug.

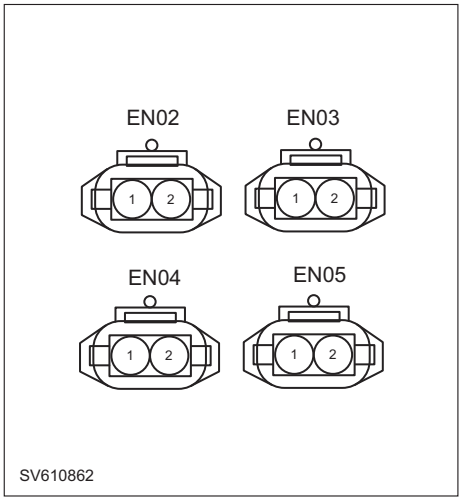
Standard value: ≈ 0Ω

5 Check whether the result is normal?

No → Repair or replace the faulted parts.

Yes ↓

Step 4	Check whether the harness of oil atomizer has grounding or power short-circuit.



1 Turn the ignition switch to OFF.

2 Disconnect the ECM harness plug.

3 Disconnect the plug of oil atomizer.

4 Measure the resistance between the plug of oil atomizer and ground.

Measuring circuit: the resistance between terminal 1 of EN02 plug and ground.

Measuring circuit: the resistance between terminal 2 of EN02 plug and ground.

Measuring circuit: the resistance between terminal 1 of EN03 plug and ground.

Measuring circuit: the resistance between terminal 2 of EN03 plug and ground.

Measuring circuit: the resistance between terminal 1 of EN04 plug and ground.

Measuring circuit: the resistance between terminal 2 of EN04 plug and ground.

Measuring circuit: the resistance between terminal 1 of EN05 plug and ground.

Measuring circuit: the resistance between terminal 2 of EN05 plug and ground.

Standard value: ∞

5 Measure the voltage between the plug of oil atomizer and ground.

Measuring circuit: the voltage between terminal 1 of EN02 plug and ground.

Measuring circuit: the voltage between terminal 2 of EN02 plug and ground.

Measuring circuit: the voltage between terminal 1 of EN03 plug and ground.

Measuring circuit: the voltage between terminal 2 of EN03 plug and ground.

Measuring circuit: the voltage between terminal 1 of EN04 plug and ground.

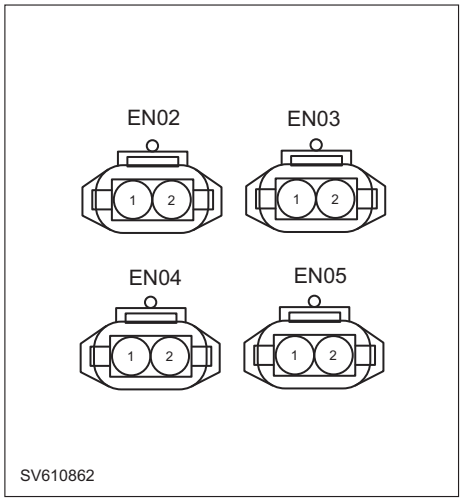
Measuring circuit: the voltage between terminal 2 of EN04 plug and ground.

Measuring circuit: the voltage between terminal 1 of EN05 plug and ground.

Measuring circuit: the voltage between terminal 2 of EN05 plug and ground.

Standard value: ≈ 0V

6 Check whether the result is normal?

	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the lines between high-end harness and low-end harness of oil atomizer (short circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the ECM harness plug. 3 Disconnect the plug of oil atomizer. 4 Measure the resistance between harness plugs of oil atomizer. Measuring circuit: the resistance between terminal 1 and terminal 2 of EN02 plug. Measuring circuit: the resistance between terminal 1 and terminal 2 of EN03 plug. Measuring circuit: the resistance between terminal 1 and terminal 2 of EN04 plug. Measuring circuit: the resistance between terminal 1 and terminal 2 of EN05 plug. **Standard value: ∞** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts .
Yes ↓	
Step 6	Check the oil atomizer.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of oil atomizer. 3 Measure the resistance between terminals of oil atomizer. Measuring circuit: the resistance between terminal 1 and terminal 2 of oil atomizer. **Standard value: 0.2~1Ω** 4 Check whether the result is normal?
	No → Replace the oil atomizer.
Next ↓	
Step 7	Complete the check and confirm whether the system works normally nor not.